Adaptively measuring quantum expectation values using the empirical Bernstein stopping rule

Bachelorarbeit

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Abstract

Quantum computing promises to be exponentially faster than classical computers, in a wide array of computational tasks. These tasks range from solving combinatorial problems or systems of equations in mathematics to simulating complex physical systems. Especially for simulating physical systems, quantum computers are thought to excel.
Although promising, a general quantum advantage over classical computers is yet to be achieved. A quantum advantage refers to a point at which quantum computers can solve complex problems that classical computers cannot solve in a finite amount of time. The current generation of quantum computers, also known as noisy intermediate scale quantum (NISQ) quantum devices, are restricted by noise. As a consequence they are limited in the number of qubits and noise Methods are developed to make the best out of the currently available hardware. Hence, these methods must deal with quantum errors, limited circuit depths, and limited qubit numbers.
A popular method are the so-called variational quantum algorithms (VQA), which employ a optimisation-based approach. VQAs employ parameterised quantum circuits, whose parameters are optimised on a classical computer.
However, VQAs face a multitude of problems, one of them being the required measurement effort in order to yield accurate results.
The goal of this thesis is to show that the measurement effort in VQAs can be reduced by the so-called empirical Bernstein stopping (EBS) algorithm. EBS provides an estimate with an accuracy guarantee Reducing the measurement effort is, however, not a unique problem of VQAs and therefore the results of this thesis will be applicable to a wider array of quantum algorithms.